

CHAPTER 6

Coordination Compounds

Introduction and Werner's Theory of Coordination Compounds

- Ethylene diaminetetraacetate (EDTA) ion is: (2021)
 - Unidentate ligand
 - Bidentate ligand with two "N" donor atoms
 - Tridentate ligand with three "N" donor atoms
 - Hexadentate ligand with four "O" and two "N" donor atoms
- The correct order of the stoichiometries of AgCl formed when AgNO_3 in excess is treated with the complexes: $\text{CoCl}_3 \cdot 6\text{NH}_3$, $\text{CoCl}_3 \cdot 5\text{NH}_3$, $\text{CoCl}_3 \cdot 4\text{NH}_3$ respectively is: (2017-Delhi)
 - 2AgCl , 3AgCl , 1AgCl
 - 1AgCl , 3AgCl , 2AgCl
 - 3AgCl , 1AgCl , 2AgCl
 - 3AgCl , 2AgCl , 1AgCl
- Cobalt(III) chloride forms several octahedral complexes with ammonia. Which of the following will not give test for chloride ions with silver nitrate at 25°C ? (2015)
 - $\text{CoCl}_3 \cdot 4\text{NH}_3$
 - $\text{CoCl}_3 \cdot 5\text{NH}_3$
 - $\text{CoCl}_3 \cdot 6\text{NH}_3$
 - $\text{CoCl}_3 \cdot 3\text{NH}_3$
- The sum of coordination number and oxidation number of the metal M in the complex $[\text{M}(\text{en})_2(\text{C}_2\text{O}_4)]\text{Cl}$ (where en is ethylenediamine) is: (2015 Re)
 - 6
 - 7
 - 8
 - 9
- The type of isomerism shown by the complex $[\text{CoCl}_2(\text{en})_2]$ is: (2018)
 - Geometrical isomerism
 - Coordination isomerism
 - Linkage isomerism
 - Ionization isomerism
- The name of complex ion, $[\text{Fe}(\text{CN})_6]^{3-}$ is: (2015 Re)
 - Hexacyanidoferrate (III) ion
 - Hexacyanoiron (III) ion
 - Hexacyanoferrate (III) ion
 - Tricyanoferrate (III) ion
- Number of possible isomers for the complex $[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$ will be: (en = ethylenediamine) (2015 Re)
 - 4
 - 2
 - 1
 - 3

Bonding in Coordination Compounds

- The order of energy absorbed which is responsible for the color of complexes (2022)
 - $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
 - $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$ and
 - $[\text{Ni}(\text{en})_3]^{2+}$
 is
 - $B > A > C$
 - $A > B > C$
 - $C > B > A$
 - $C > A > B$
- Match List-I with List-II. (2021)

List-I		List-II	
(A)	$[\text{Fe}(\text{CN})_6]^{3-}$	(i)	5.92 BM
(B)	$[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$	(ii)	0 BM
(C)	$[\text{Fe}(\text{CN})_6]^{4-}$	(iii)	4.90 BM
(D)	$[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$	(iv)	1.73 BM

Choose the correct answer from the options given below.

- The IUPAC name of the complex- (2022)
 $[\text{Ag}(\text{H}_2\text{O})_2][\text{Ag}(\text{CN})_2]$ is:
 - diaquasilver(I) dicyanidoargentate(I)
 - dicyanidosilver(II) diaquaargentate(II)
 - diaquasilver(II) dicyanidoargentate(II)
 - dicyanidosilver(I) diaquaargentate(I)

- A-ii B-iv C-iii D-i
- A-i B-iii C-iv D-ii
- A-iv B-i C-ii D-iii
- A-iv B-ii C-i D-iii

Nomenclature & Isomerism in Coordination Compounds

11. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds? (2020)
- $\text{SCN}^- < \text{F}^- < \text{CN}^- < \text{C}_2\text{O}_4^{2-}$
 - $\text{F}^- < \text{SCN}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
 - $\text{CN}^- < \text{C}_2\text{O}_4^{2-} < \text{SCN}^- < \text{F}^-$
 - $\text{SCN}^- < \text{F}^- < \text{C}_2\text{O}_4^{2-} < \text{CN}^-$
12. Match the coordination number and type of hybridisation with distribution of hybrid orbitals in space based on Valence bond theory. (2020-Covid)

Coordination number and type of hybridisation		Distribution of hybrid orbitals in space	
(A)	4, sp^3	(i)	Trigonal bipyramidal
(B)	4, dsp^2	(ii)	Octahedral
(C)	5, sp^3d	(iii)	Tetrahedral
(D)	6, d^2sp^3	(iv)	Square planar

Select the correct option:

- A-iii B-iv C-i D-ii
 - A-iv B-i C-ii D-iii
 - A-iii B-i C-iv D-ii
 - A-ii B-iii C-iv D-i
13. What is the correct electronic configuration of the central atom in $\text{K}_4[\text{Fe}(\text{CN})_6]$ based on crystal field theory? (2019)
- $t_{2g}^4 e_g^2$
 - $t_{2g}^6 e_g^0$
 - $e^3 t_{2g}^3$
 - $e^4 t_{2g}^2$
14. The geometry and magnetic behaviour of the complex $[\text{Ni}(\text{CO})_4]$ are? (2018)
- Square planar geometry and diamagnetic
 - Tetrahedral geometry and diamagnetic
 - Tetrahedral geometry and paramagnetic
 - Square planar geometry and paramagnetic
15. Correct increasing order for the wavelengths of absorption in the visible region for the complexes of Co^{3+} is: (2017-Delhi)
- $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 - $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 - $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 - $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, $[\text{Co}(\text{en})_3]^{3+}$
16. Pick out the correct statement with respect to $[\text{Mn}(\text{CN})_6]^{3-}$ (2017-Delhi)
- It is dsp^2 hybridised and square planar
 - It is sp^3d^2 hybridised and octahedral
 - It is sp^3d^2 hybridised and tetrahedral
 - It is d^2sp^3 hybridised and octahedral
17. The correct increasing order of trans-effect of the following species is: (2016-II)
- $\text{NH}_3 > \text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^-$
 - $\text{CN}^- > \text{C}_6\text{H}_5^- > \text{Br}^- > \text{NH}_3$
 - $\text{Br}^- > \text{CN}^- > \text{NH}_3 > \text{C}_6\text{H}_5^-$
 - $\text{CN}^- > \text{Br}^- > \text{C}_6\text{H}_5^- > \text{NH}_3$

18. Jahn-Teller effect is not observed in high spin complexes of: [OS] (2016-II)

- d^7
- d^8
- d^4
- d^9

19. The hybridisation involved in complex $[\text{Ni}(\text{CN})_4]^{2-}$ is (Atomic Number Ni = 28) (2015 Re)

- d^2sp^3
- dsp^2
- sp^3
- d^2sp^2

20. Which of these statements about $[\text{Co}(\text{CN})_6]^{3-}$ is true? (2015)

- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a low-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has four unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a high-spin configuration
- $[\text{Co}(\text{CN})_6]^{3-}$ has no unpaired electrons and will be in a low-spin configuration

21. Among the following complexes the one which shows zero crystal field stabilization energy (CFSE) is: (2014)

- $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
- $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
- $[\text{Mn}(\text{H}_2\text{O})_6]^{3+}$

22. A magnetic moment of 1.73 BM will be shown by one among the following: (2013)

- $[\text{Cu}(\text{NH}_3)_4]^{2+}$
- $[\text{Ni}(\text{CN})_4]^{2-}$
- TiCl_4
- $[\text{CoCl}_6]^{4-}$

Bonding in Metal Carbonyls

23. Iron carbonyl, $\text{Fe}(\text{CO})_5$ is (2018)

- Tetranuclear
- Mononuclear
- Dinuclear
- Trinuclear

24. Which of the following has longest C-O bond length? (Free C-O bond length in CO is 1.128 Å) (2016 - I)

- $[\text{Mn}(\text{CO})_6]^+$
- $\text{Ni}(\text{CO})_4$
- $[\text{Co}(\text{CO})_4]^-$
- $[\text{Fe}(\text{CO})_4]^{2-}$

Applications of Coordination Compounds

25. Which of the following complexes is used to be as an anticancer agent? (2014)

- cis- $[\text{PtCl}_2(\text{NH}_3)_2]$
- cis- $\text{K}_2[\text{PtCl}_2\text{Br}_2]$
- $\text{Na}_2[\text{CoCl}_4]$
- mer- $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
d	d	d	d	a	a	a	d	d	c	d	a	b	b	b	d	b
18	19	20	21	22	23	24	25									
b	b	d	a	a	b	d	a									

